Spectral Knowledge Transfer for Remote Sensing Change Detection

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***Abstract*— Change detection (CD) in multispectral remote sensing (RS) imagery suffers from low spectral resolution which can lead to degraded recognition of change information from land cover objects. Considering that natural hyperspectral imagery (HSI) is much higher in spectral resolution and more accessible, using it to enhance the spectral information of RS multispectral imagery for CD can improve performance. To achieve this, we propose a spectral knowledge transfer (SKT) framework to allow the creation of pseudo-hyperspectral RS images from the available RS multispectral ones without the need for the real pairs of RS multispectral and hyperspectral images, typically required by existing RS spectral enhancement methods. Specifically, an autoencoder is first trained based on the available pairs of natural HSI and its multispectral counterparts and then calibrated via the available RS multispectral images. The finally obtained decoder module is used to generate the pseudo-hyperspectral image from an input RS multispectral image. We further propose a multispectrum collaborative CD (MCCD) framework that leverages both the real multispectral images and the pseudo hyperspectral images generated from them in a collaborative way to achieve performance improvement. Extensive experiments on two large-scale RS CD datasets and eight existing deep learning-based CD methods demonstrate the stronger efficacy of the proposed method.**

***Index Terms*— Change detection (CD), data augmentation, deep learning, very-high-resolution (VHR) remote sensing (RS) images.**

I. Introduction

CHANGE detection (CD) can assess the impact of geographical events over specific areas since it aims to discover meaningful land cover changes spatially co-registered. Bi-temporal remote sensing (RS) images [1]. It has been extensively used in various RS applications, e.g., urban land use investigation [2], disaster assessment [3], and forest resource monitoring [4]. Based on actual needs, it can be performed over multiple sources of data, e.g., synthetic aperture radar imagery [5], [6], very-high-resolution (VHR) multispectral imagery [7], and heterogeneous bi-temporal images from different sensors [8], [9].

A. Background

With the development of sensors, massive RS data are acquired every moment, which greatly promotes the progress in CD techniques [10]. Among them, multispectral images play a significant role [11], since they often have higher spatial resolution, which facilitates the recognition of the land cover objects with smaller sizes or more details. RGB imagery is the most easily obtained multispectral data, which can be captured by both aerial and satellite sensors. Based on these data, many CD methods have been invented to better acquire change information. Recently, the majority of state-of-the-art CD methods have been built over deep neural networks [5], [12], [13], e.g., fully convolutional network (FCN) [14] and ResNet [15]. Thanks to the outstanding image recognition ability of these networks, the CD methods based on them have achieved great success in multispectral CD tasks. For example, Daudt et al. [16] proposed three different FCN-based architectures to conduct CD in multispectral images, which are used by many following works. Fang et al. [17] proposed a densely connected Siamese network (SNU-Net) to leverage the change information from both deep and shallow layers for better CD performance in VHR multispectral images. They also introduced an ensemble channel attention module for finer feature representation. Based on the success of the self-attention mechanism [18], there are also many works trying to utilize self-attention techniques to acquire finer change information in multispectral CD tasks. For instance, a spectral–temporal attention-based network (STA-Net) was designed in [19] to capture spatial–temporal dependency by introducing a CD self-attention mechanism. Moreover, Chen et al. [20] proposed a bi-temporal transformer (BIT) that directly uses the transformer architecture in multispectral CD.

B. Motivation

Though these methods have been making progress for multispectral CD, the performance is strictly limited by insufficient

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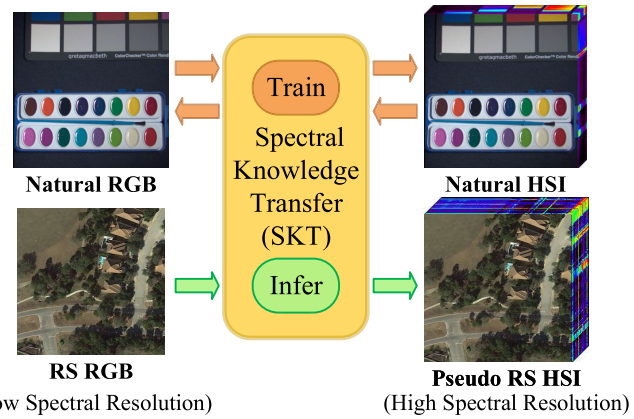


Fig. 1. Proposed SKT framework aims to employ the abundant R2H mapping knowledge in natural images to enhance the spectral information of multispectral remote sensed imagery. The HSI and RS indicate hyperspectral imagery and remote sensing, respectively.

spectral resolution, which can lead to graver problems of "different objects with the same spectrum" and "same object with a different spectrum." It is also difficult to directly turn to hyperspectral CD. On the one hand, the remotely sensed hyperspectral bi-temporal images are more difficult to obtain [21]. On the other hand, in addition to the difficulty of obtaining RS hyperspectral data, RS hyperspectral imagery (HSI) suffers from low spatial resolution issues due to the nature of imaging facilities. Besides, hyperspectral data are much richer in spectral information, which allows related CD methods to use fewer training samples to reach better performance. This enables more unsupervised and semisupervised hyperspectral CD methods [22]. Based on this fact, enhancing the spectral information of bi-temporal multispectral images to build the imagery with both high spatial and spectral resolution can potentially benefit the CD performance of existing methods.

To achieve this, we propose a spectral knowledge transfer (SKT) framework to leverage hyperspectral images to enhance the spectrum of multispectral images, as shown in Fig. 1. Different from existing similar works, we utilize natural hyperspectral images rather than remote-sensed HSI. On the one hand, natural hyperspectral images are easier to acquire compared to remote-sensed hyperspectral images. On the other hand, remote-sensed HSI always has a mixed spectrum due to its low spatial resolution, which can cause distortion in enhanced multispectral images. The SKT can generate bi-temporal pseudo-imagery with both high spectral and spatial resolution. Based on enhanced bi-temporal images, we design a new CD framework, multispectrum collaborative CD (MCCD), to fully leverage the origin images and enhanced images to obtain better CD performance over existing state-of-the-art methods. Our contributions can be summarized in four folds as follows.

1) We present the SKT strategy to transfer spectral information from natural HSI to multispectral remote sensed imagery, which can generate pseudo-imagery with both high spectral and spatial resolution. The corresponding experimental results suggest that it can transfer spectral knowledge from natural hyperspectral images to remotely sensed multispectral imagery to expand the spectrum of used data for better performance in CD tasks.
2) We propose an MCCD framework that both uses real multispectral imagery and generated pseudo-HSI. It can be implemented over a wide range of existing deep learning-based CD methods. The performance improvement is validated via extensive experiments.

II. Related Works

A. CD on Multispectral Imagery

As the skyrocketing volume of multispectral images keeps coming from aerial and satellite sensors, the multispectral CD has become one of the research highlights in the RS [23] and computer vision fields [24], [25]. In this article, we divided these CD methods into patch-based and image-based methods according to their basic input units.

When neural networks are introduced into a CD at the early stage, there are limitations for the network-based CD methods to efficiently process the bi-temporal RS images with higher resolution, since the early neural networks usually have a restricted scale of learnable parameters which leads to the limited ability of feature extraction and CD [26]. Under this circumstance, the input bi-temporal images are divided into smaller paired image patches to lower the burden of the neural networks [27]. These methods employ extracted patch-wise change information to build a pixel-wise binary change map [28]. For example, Shu et al. [29] proposed a multilayer fusion network to extract the patch-wise feature of bi-temporal SAR images and predict the changed areas patch by patch. Besides, super-pixel-based CD methods [30] can also be seen as patch-based CD methods with heterogeneous patches. Different from other patch-based CD methods, super-pixel-based CD methods first divided the bi-temporal image pairs into various super-pixels according to pixel-wise similarity [31]. Then, binary change maps are generated super-pixel by super-pixel by evaluating the change information of each super-pixel. For example, Zheng et al. [32] segmented bi-temporal images into individual homogeneous regions separately to deal with cross-resolution imagery and built corresponding difference maps. Then, the final binary change map is generated by fusing cross-resolution difference maps. For instance, Lv et al. [33] first segment the bi-temporal image pair into multiscale super-pixels according to the pixel-wise difference intensity. Then, a denoising autoencoder [34], [35] was employed to detect the change information of vectorized super-pixels. The patch-based CD methods achieved good performance for multispectral RS imagery at that time. When lacking training samples, patch-based methods are acceptable frameworks for CD. However, the optimal patch size is relatively small and difficult to determine, which limits the use of spatial information and the extension of these methods. To deal with this problem, a feature-regularized mask DeepLab (FRM-DeepLab) is proposed in [36]. However, the patch-wise CD framework of these methods stops them from reaching higher efficiency when

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dealing with large-scale VHR RS imagery, e.g., WHU [37], SYSU-CD [38], and CDD [39] datasets. On the one hand, patch-based detection often fails to acquire global and contextual information in the scenes. On the other hand, VHR large-scale datasets have to be divided into massive patches if using patch-based methods, which makes them unavailable under this circumstance.

With the advance of deep learning techniques [15], neural networks can have many more hidden layers with a skyrocketing number of learnable parameters [40], which greatly improve the performance for feature extraction and pattern recognition [41]. This also further promotes the advance of RS image processing applications [42] like scene classification [43], segmentation [44], and CD [45]. As for the CD in multispectral imagery, many convolutional neural network-based methods have achieved an image-to-image pipeline with higher efficiency compared to patch-based methods. The convolutional neural networks with larger scale, e.g., U-Net [46] and ConvNeXt [47], can be utilized to effectively deal with large-scale VHR CD tasks in an end-to-end framework, which gives birth to many image-based CD methods. For example, Zheng et al. [48] proposed a new Siamese encoder–decoder convolutional network that aims to enhance high-frequency information for better CD performance in VHR multispectral imagery. Lei et al. [49] invented the difference enhancement and spatial–spectral nonlocal modules to lower the computational cost and improve the performance of Siamese CD networks. Besides, Zhang et al. [7] introduced a super-pixel enhancement strategy into convolutional neural networks for multispectral CD tasks, which suppresses latent noise and preserves object edges. In summary, the image-based methods have further promoted the CD in large-scale VHR multispectral remote sensed imagery. However, the requirements of CD tasks also have increased and become more varied. The CD tasks often do not care about all kinds of land cover changes. The CD methods have to recognize useful land cover changes according to the guidance of ground truth. For example, the building CD tasks only focus on the changes in built-up areas [50]. Besides, with the increase of spatial resolution, the VHR RS multispectral imagery usually contains much more details with small land cover objects, thus making the depicted scenes much more complex, which challenges the CD performance of image-based methods [51].

To counter the aforementioned challenges, attention mechanisms [52], [53] have been focused on and utilized by researchers in the RS field. It is found that attention mechanisms can help the deep learning-based methods selectively focus on relevant objects according to the need of tasks [54]. In CD tasks, attention mechanisms can help the methods ignore irrelevant land cover changes and enhance the relevant change information in feature level [55]. Based on this fact, many image-based CD methods are built over the attention mechanisms, e.g., spatial attention [56], channel attention [57], and self-attention [58]. For example, in [59], a simple but effective channel attention module [60] is employed to enhance spectral bands with rich information for CD. Besides, in [61], a dual-way attention mechanism including spatial and spectral attention is proposed to deal with CD in HSI. With the help of varied attention mechanisms, these CD methods can achieve better recognition of the spectral and semantic differences of varied land cover objects. However, their CD performance improvement is still inhibited by the low spectral resolution and insufficient spectral knowledge of multispectral imagery, which can lead to poor recognition of various land cover objects.

In this work, we make efforts to improve the spectral resolution of RS multispectral imagery with the help of heterogeneous natural imagery, which has the potential to directly overcome the aforementioned limitations in a different way compared to existing CD methods, since the proposed SKT and MCCD can offer rich extra spectral information to greatly facilitate the recognition of varied land cover objects for CD in multispectral imagery.

B. Spectral Resolution Enhancement on Remote Sensed Imagery

Compared to multispectral images, hyperspectral remotely sensed imagery is more difficult to obtain due to the narrow swarth of corresponding sensors [21]. Many works have been made on acquiring HSI from multispectral images via modeling the relationship between their spectra. For example, Sun et al. [21] used linear relationships to describe the transformation from multispectra to hyperspectral, which generates a set of matrices changing pixel-wise multispectral vectors to hyperspectral ones. Besides, Arun et al. [62] utilized a convolutional neural network to better model the spatial–spectral prior for a more precise reconstruction from multispectral images to hyperspectral ones. Apart from that, high spatial resolution multispectral imagery is also utilized to guide the generation of HSI. For example, in [63] and [64], high spatial multispectral images are employed to guide the spectral and spatial reconstruction of hyperspectral images in unsupervised and semisupervised frameworks, respectively. Conventional spectral enhancement methods mainly face two challenges. First, most of them rely on prior knowledge such as the spectral response of sensors to generate high-fidelity hyperspectral images [65]. Second, these methods often need paired RS multispectral and hyperspectral remotely sensed imagery, which is a narrow source of data. Unlike conventional works, we transfer spectral knowledge learned from natural RGB-hyperspectral image pairs to remotely sensed multispectral imagery without sensor priors, which makes the SKT a more flexible framework. Moreover, we test the spectral enhancement performance over CD tasks rather than indicators like signal-to-noise ratio (SNR), which further promotes the application of enhanced imagery.

III. Methods

In this section, the proposed method is introduced in detail in two main parts. 1) A learning strategy, SKT, is proposed to reliably transfer spectral information from natural HSI to remotely sensed multispectral images. 2) A new CD framework, MCCD, is designed for fully leveraging generated pseudo-hyperspectral data and origin multispectral RS data to improve the performance of existing deep learning-based CD methods.

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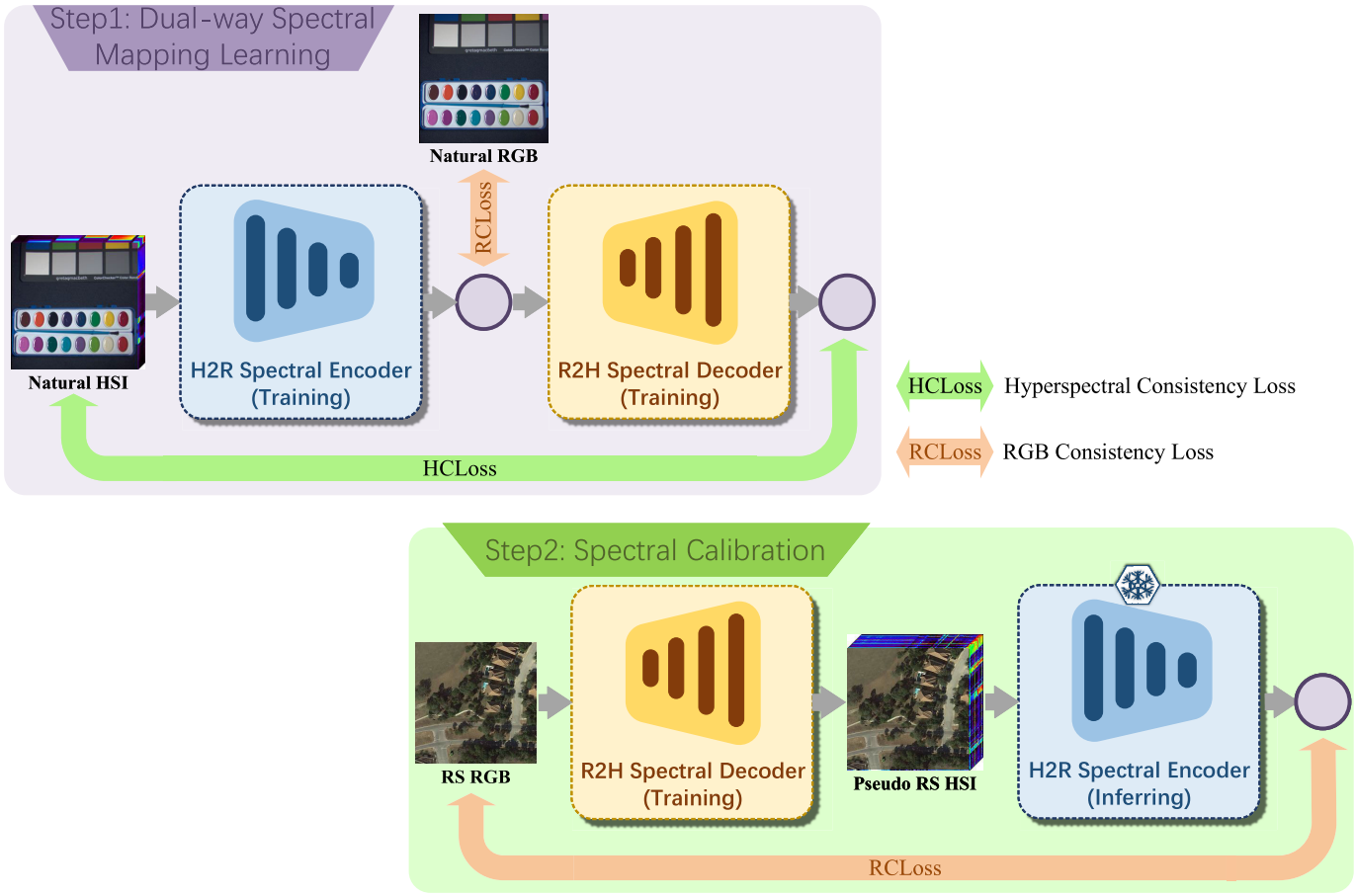


Fig. 2. Brief illustration of the proposed SKT. The snowflake means that the parameters of the model are frozen and stop updating.

A. Spectral Knowledge Transfer

1) Dual-Way Spectral Mapping Learning in Natural Imagery: Considering that the multispectral imagery for CD is often low in spectral resolution, which can lead to the corrupted cognition of different land cover objects, we try to enhance the spectral information of the RS multispectral imagery directly. However, the paired RS multispectral and HSI are often hard to acquire, which stops most conventional spectral enhancement methods from collecting sufficient data for supervised learning. To bypass this obstacle, we utilize a natural spectral reconstruction dataset to learn the natural spectral knowledge which can be transferred to RS multispectral imagery for higher spectral resolution. To transfer natural spectral knowledge, we first need to model the mapping from RGB to hyperspectral, which is a highly ill-posed problem. To alleviate this problem, we employ the encoder–decoder architecture to learn the dual-way spectral mappings, i.e., hyperspectra-to-RGB (H2R) and RGB-to-hyperspectra (R2H), as shown in Fig. 2. In this procedure, the natural HSI is first input into the H2R spectral encoder, compressing the number of spectrum bands to 3, which can be presented as

$$\widetilde{x}_{\mathrm{RGB}} = f_{\mathrm{H2R}}(x_{\mathrm{HSI}}) \tag{1}$$

where x_{HSI} denotes the real natural HSI, $f_{\mathrm{H2R}}(\cdot)$ represents the H2R spectral encoder, and $\widetilde{x}_{\mathrm{RGB}}$ is the reconstructed RGB imagery. Notably, the H2R spectral encoder contains only 1×1 convolutional layers with batch normalization and ReLU, which does not manipulate the spatial size of input images, since we try to avoid the distortion in spatial dimension, which can be poison for CD. To be more specific, the H2R spectral encoder has three 1×1 convolutional layers with batch normalization and ReLU. The first layer changes channel size from 31 to 16. The second layer changes channel size from 16 to 8. The last layer changes channel size from 8 to 3. Finally, a sigmoid layer is utilized to acquire stable output. The R2H spectral decoder is the reverse version of the H2R spectral encoder, which gradually increases the channel size.

The next step is to reconstruct the HSI via the R2H spectral decoder, which can be written as

$$\widetilde{x}_{\mathrm{HSI}} = f_{\mathrm{R2H}}\left(\widetilde{x}_{\mathrm{RGB}}\right) \tag{2}$$

where $f_{\mathrm{R2H}}(\cdot)$ and $\widetilde{x}_{\mathrm{HSI}}$ indicate the R2H spectral decoder and reconstructed natural HSI, respectively. It is notable that the proposed R2H spectral decoder is symmetrical to the H2R spectral encoder to avoid spatial distortion.

To learn the H2R and R2H mappings inside the natural imagery simultaneously, we employ two loss functions, RGB consistency loss (RCLoss) and HSI consistency loss (HCLoss), to ensure the reconstructed RGB and HSI imagery is similar

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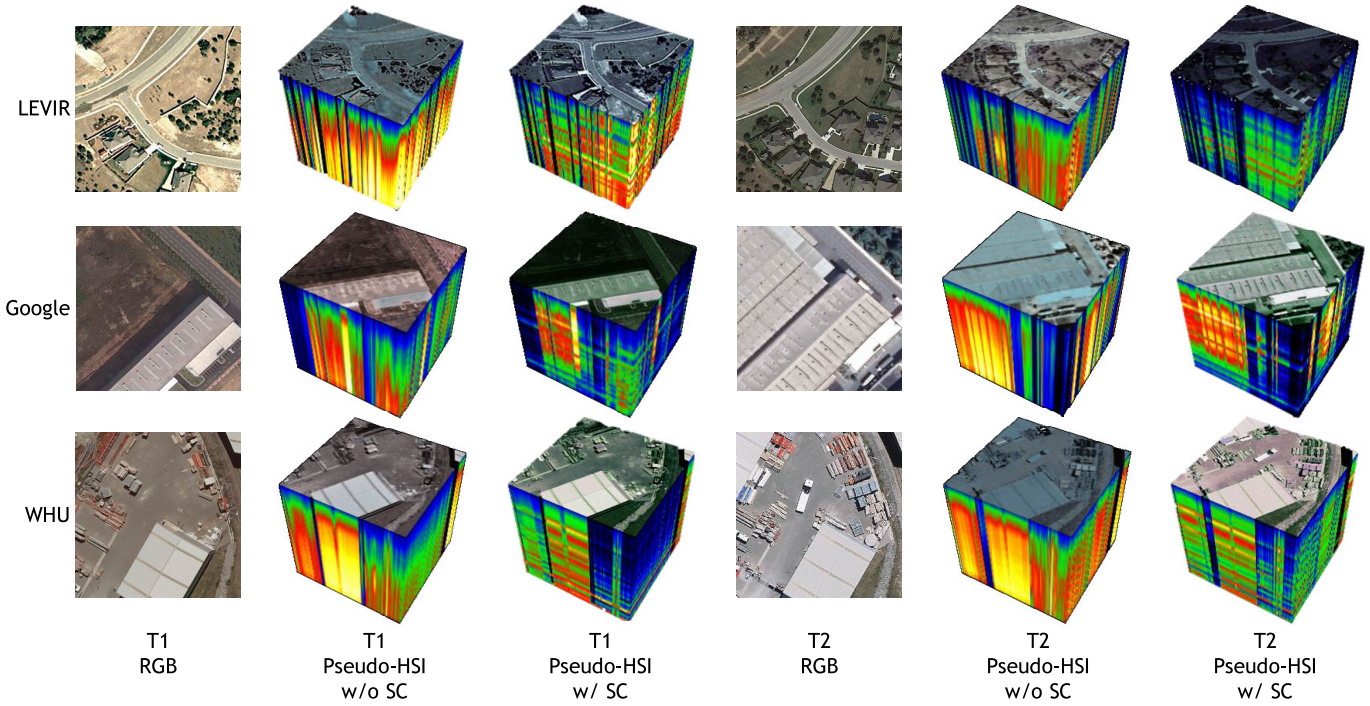


Fig. 3. Some pseudo-HSI before and after spectral calibration generated from three used large-scale datasets. The SC stands for spectral calibration. The w/ stands for with, and w/o stands for without.

enough to the real ones in spectral dimension, which can be denoted as

$$\mathcal{L}_{\mathrm{DSML}} = \mathcal{L}_{\mathrm{MSE}}\left(\widetilde{x}_{\mathrm{RGB}}, x_{\mathrm{RGB}}\right) + \mathcal{L}_{\mathrm{MSE}}\left(\widetilde{x}_{\mathrm{HSI}}, x_{\mathrm{HSI}}\right) \quad (3)$$

where $\mathcal{L}_{\mathrm{DSML}}$ represents the total loss for dual-way spectral mapping learning and $\mathcal{L}_{\mathrm{MSE}}(\cdot)$ denotes the mean square error loss function. In this stage, the training ends when the $\mathcal{L}_{\mathrm{DSML}}$ stops dropping. With the help of the symmetrical encoder–decoder architecture, we continuously model the H2R and R2H spectral mapping, which avoids learning R2H mapping in an isolated process, since it is highly ill-posed.

In summary, an encoder–decoder architecture is employed to obtain the natural spectral information of H2R and R2H. With the R2H spectral information of natural imagery acquired, it is possible to transform bi-temporal RS multispectral images to hyperspectral images for better CD performance, which has been validated in the experiments behind. However, considering that the remotely sensed imagery differs from the natural imagery captured by the sensors on the ground, the learned natural R2H spectral mapping needs to be further adjusted.

2) Spectral Calibration in Remotely Sensed Imagery: Through dual-way spectral mapping learning, we have acquired the R2H mapping over the natural imagery. However, there is still a latent problem barricading the acquisition of workable RS pseudo-HSI. That is, RS HSI and natural HSI differ greatly in imaging conditions and sensors, which may lead to irrational pseudo-HSI. Based on this fact, it is necessary to calibrate the spectral mapping by using real RS multispectral imagery. If not, the ill bi-temporal pseudo-HSI transformed from real multispectral RS imagery can be poison for CD performance. To achieve this, we freeze the parameters of the H2R spectral encoder, and employ it to help with the calibration of the R2H spectral decoder over multispectral RS imagery, which is inspired by the style transfer strategy of

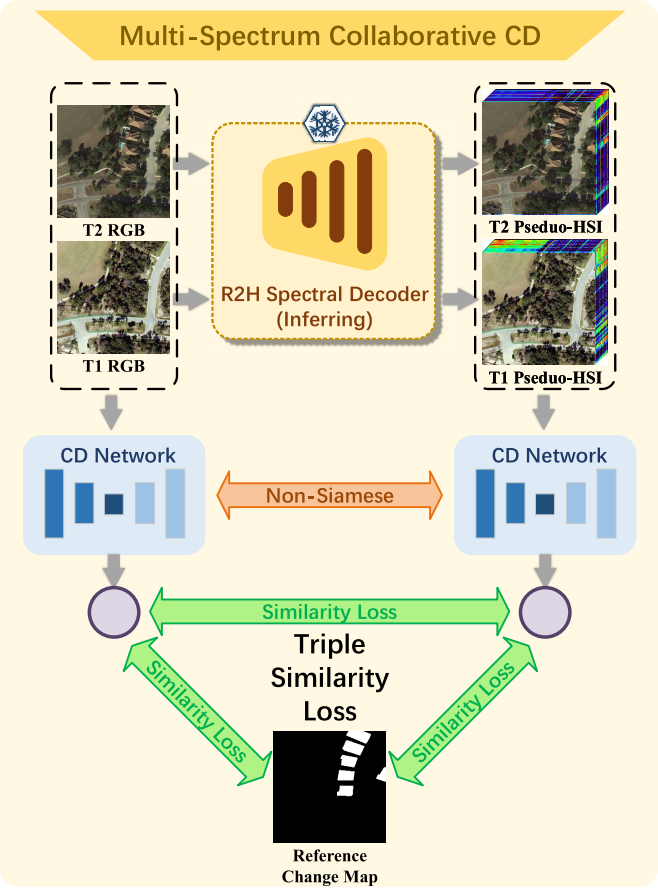


Fig. 4. Illustration of the proposed MCCD. The snowflake means that the parameters of the model are frozen and stop updating.

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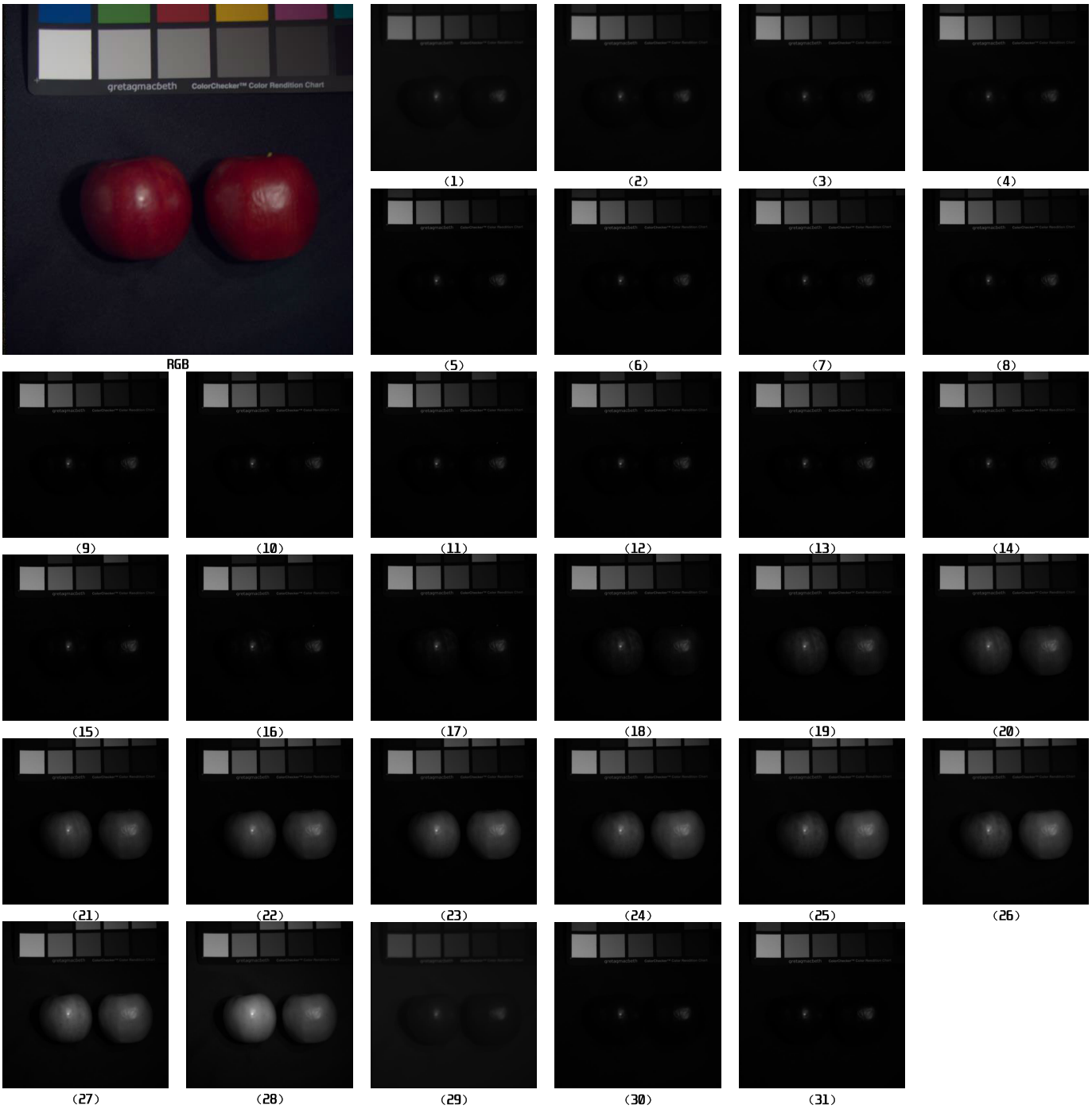


Fig. 5. Example from CAVE spectral reconstruction dataset. This set of images consists of an RGB image and corresponding 31 images including 31 spectral bands of the corresponding HSI, which depict a real and a fake apple. It can be seen that even though they are similar in the RGB image, they still differ greatly in some bands of the hyperspectral image, e.g., (25) and (28). This indicates that the generated pseudo-HSI may help the CD methods acquire better cognition of the varied land cover objects which are similar in multispectral imagery.

cycle-consistent adversarial network [66]. The detailed process of spectral calibration is illustrated in Fig. 2. First, real RS RGB imagery is input into the R2H spectral decoder trained in dual-way spectral learning, which can be represented as

$$\widetilde{\mathbb{X}}_{\mathrm{HSI}} = f_{\mathrm{R2H}}(\mathbb{X}_{\mathrm{RGB}}) \tag{4}$$

where $\mathbb{X}_{\mathrm{RGB}}$ and $\widetilde{\mathbb{X}}_{\mathrm{HSI}}$ denote the real RS multispectral image and generated RS pseudo-HSI, respectively. To evaluate and recalibrate the generated HSI, we utilize the frozen H2R spectral encoder to remap the pseudo-HSI to RGB bands, which can be denoted as

$$\widetilde{\mathbb{X}}_{\mathrm{RGB}} = \bar{f}_{\mathrm{H2R}}\left(\widetilde{\mathbb{X}}_{\mathrm{HSI}}\right) \tag{5}$$

where $\widetilde{\mathbb{X}}_{\mathrm{RGB}}$ and $\bar{f}_{\mathrm{H2R}}(\cdot)$ denote reconstructed RGB and the H2R spectral encoder with fixed parameters. To register the R2H spectral mapping, we employ a mean square error loss

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Fig. 6. Some examples from LEVIR, Google, and WHU large-scale VHR multispectral CD datasets.

TABLE I
QUANTITATIVE RESULTS (IN %) OVER STATE-OF-THE-ART BASELINES

Methods		Data Sets											
		LEVIR				Google				WHU			
		F1-Score	**mIoU**	Precision	Recall	**F1-Score**	**mIoU**	Precision	Recall	**F1-Score**	**mIoU**	Precision	Recall
Backbones	FCN	86.96	87.82	88.40	85.57	78.70	80.09	81.59	**76.00**	85.18	85.46	88.88	81.78
	- w/ MCCD	**87.94**(+0.98)	**88.65**(+0.83)	**89.85**	**86.11**	**79.68**(+0.98)	**80.90**(+0.81)	**83.89**	75.87	**88.07**(+2.89)	**88.03**(+2.57)	**92.74**	**83.84**
	ResNet	88.45	89.08	90.26	86.70	81.21	82.01	80.55	**81.89**	85.66	85.85	87.48	83.93
	- w/ MCCD	**89.35**(+0.90)	**89.86**(+0.78)	**91.98**	**86.88**	**82.58**(+1.37)	**83.27**(+1.26)	**87.54**	78.15	**90.70**(+5.07)	**90.46**(+4.61)	**94.97**	**86.79**
CD methods	FC-EF	83.00	84.63	83.36	**82.65**	65.96	71.30	77.69	57.31	84.21	84.59	86.11	82.39
	- w/ MCCD	**83.94**(+0.94)	**85.39**(+0.76)	**87.99**	80.24	**68.32**(+2.36)	**72.93**(+1.63)	**83.07**	**58.02**	**84.99**(+0.78)	**85.25**(+0.66)	**86.32**	**83.70**
	FC-Siam-Conc	84.76	86.08	**93.77**	77.32	66.83	71.98	**82.39**	56.21	66.56	69.69	52.76	90.16
	- w/ MCCD	**87.22**(+2.46)	**88.05**(+1.97)	89.87	**84.73**	**67.82**(+0.99)	**72.55**(+0.57)	80.71	**58.48**	**68.38**(+1.82)	**71.13**(+1.44)	**55.01**	**90.36**
	FC-Siam-Diff	83.76	85.28	90.75	77.77	62.33	69.22	**80.29**	50.94	71.08	73.42	59.54	**88.16**
	- w/ MCCD	**85.42**(+1.66)	**86.60**(+1.32)	**93.20**	**78.83**	**62.83**(+0.50)	**69.33**(+0.11)	74.78	**54.18**	**73.19**(+2.11)	**75.31**(+1.89)	**64.69**	84.27
	SNU-Net	86.43	87.40	90.38	82.81	49.58	60.84	49.17	**50.01**	74.30	76.25	67.07	83.29
	- w/ MCCD	**88.03**(+1.60)	**88.73**(+1.33)	**90.71**	**85.50**	**51.61**(+2.03)	**62.78**(+1.94)	**61.22**	44.61	**75.81**(+1.51)	**77.47**(+1.22)	**69.55**	**83.32**
	STA-Net	83.30	84.77	76.69	**91.15**	61.10	68.18	70.75	**53.76**	84.91	85.09	82.47	87.51
	- w/ MCCD	**85.07**(+1.77)	**86.28**(+1.51)	**86.81**	83.39	**62.51**(+1.41)	**69.26**(+1.08)	**78.31**	52.01	**87.05**(+2.14)	**87.01**(+1.92)	**85.53**	**88.63**
	BIT	88.48	89.11	**90.35**	86.69	78.96	80.46	**88.66**	71.17	84.30	84.51	**79.80**	89.35
	- w/ MCCD	**88.66**(+0.18)	**89.25**(+0.14)	89.06	**88.28**	**80.64**(+1.68)	**81.71**(+1.25)	86.74	**75.33**	**84.66**(+0.36)	**84.80**(+0.29)	79.63	**90.37**
	ChangeFormer	88.54	89.14	87.79	**89.30**	78.29	79.89	**85.65**	72.10	83.78	84.00	77.68	**90.92**
	- w/ MCCD	**88.72**(+0.18)	**89.31**(+0.17)	**90.53**	86.97	**81.49**(+3.20)	**82.35**(+2.46)	85.07	**78.20**	**86.09**(+2.31)	**86.10**(+2.10)	**82.50**	90.00

function to compare the $\mathbb{X}_{\mathrm{RGB}}$ with $\widetilde{\mathbb{X}}_{\mathrm{RGB}}$ and make them be inconsistent enough with each other, which can be represented as

$$\mathcal{L}_{\mathrm{SC}} = \mathcal{L}_{\mathrm{MSE}}\left(\widetilde{\mathbb{X}}_{\mathrm{RGB}}, \mathbb{X}_{\mathrm{RGB}}\right) \tag{6}$$

where $\mathcal{L}_{\mathrm{SC}}$ is the total loss for spectral calibration in remotely sensed imagery.

In summary, it is difficult to directly evaluate the quality of generated RS pseudo-HSI, since there does not really exist a paired real RS HSI for most multispectral RS imagery. Based on this fact, we employ learned natural H2R spectral mapping to evaluate and guide the generation of pseudo-HSI indirectly. Once the RGB imagery transformed from pseudo-HSI imagery is similar or almost identical to the source RGB imagery, the natural R2H, and H2R mapping knowledge can be seen as rebuilt over RS imagery, which may indicate the applicability of SKT. However, whether the generated pseudo-HSI is rational and informative or not still awaits to be tested in the CD tasks. Once the generated bi-temporal can support the existing CD methods to achieve higher performance, it can be seen as a success of the proposed SKT, since the final purpose is to facilitate the complex CD tasks in VHR multispectral imagery. As shown in Fig. 3, we visualized some generated pseudo-hyperspectral data before and after spectral calibration. It can be seen from the figure that the spectral information becomes more complicated after spectral calibration. It suggests that the pseudo-HSI after spectral calibration can be more informative, which can better support the CD tasks in the proposed MCCD.

B. Multispectrum Collaborative CD

Through the proposed SKT, the bi-temporal RS pseudo-HSI, which has been preliminarily verified by spectral calibration, is obtained. With paired pseudo-HSI acquired, multispectral

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Algorithm 1 Workflow of MCCD

Input:
Bi-temporal real multispectral RS data $\mathcal{T}^1_{RGB}$, $\mathcal{T}^2_{RGB}$;
Output:
Final binary change map CM;

```
Generate bi-temporal pseudo-HSI T^1_HSI, T^2_HSI by trained and calibrated R2H spectral decoder f_R2H(·);
Utilize non-Siamese CD networks F_RGB(·) and F_HSI(·) to get the change maps CM_RGB and CM_HSI, respectively;
if Training then
    Use the reference change map (ground truth) CM_Ref to build the triple similarity loss function for training the networks in the proposed MCCD;
else {Inferring}
    CM = CM_HSI;
end if
return Output final binary change map CM;
```

CD methods can be informed with higher spectral resolution. A new CD framework called MCCD, which can suit a wide range of deep learning-based CD methods, is proposed to utilize both generated pseudo-HSI and real multispectral bi-temporal RS images. Though using more data does need more computational resources, it can be beneficial for two reasons. First, more data is often good for data-driven deep learning-based methods. Second, real multispectral data can adjust the supervised learning in MCCD if a few corrupted pseudo-HSIs occur in the framework. To fully leverage the real multispectral RS imagery and generated pseudo-HSI, we use them collaboratively in the proposed CD framework. To achieve this, we employ two non-Siamese networks from the same method to build the CD framework, as shown in Fig. 4. First, the bi-temporal pseudo-HSI is acquired by a trained R2H spectral decoder, which can be shown as

$$\begin{aligned} \mathcal{T}^1_{\mathrm{HSI}} &= \bar{f}_{\mathrm{R2H}}\left(\mathcal{T}^1_{\mathrm{RGB}}\right) \\ \mathcal{T}^2_{\mathrm{HSI}} &= \bar{f}_{\mathrm{R2H}}\left(\mathcal{T}^2_{\mathrm{RGB}}\right) \end{aligned} \tag{7}$$

where $\bar{f}_{\mathrm{R2H}}(\cdot)$ denotes the learned R2H decoder calibrated by real multispectral RS data. $\mathcal{T}^1_{\mathrm{RGB}}$, $\mathcal{T}^2_{\mathrm{RGB}}$ and $\mathcal{T}^1_{\mathrm{HSI}}$, $\mathcal{T}^2_{\mathrm{HSI}}$ represent the bi-temporal real multispectral data and pseudo hyperspectral data, respectively. Then the multispectrum RS data are, respectively, input into the CD networks of two different branches, which can be denoted as

$$\begin{aligned} \mathrm{CM}_{\mathrm{RGB}} &= \mathcal{F}_{\mathrm{RGB}}\left(\mathcal{T}^1_{\mathrm{RGB}}, \mathcal{T}^2_{\mathrm{RGB}}\right) \\ \mathrm{CM}_{\mathrm{HSI}} &= \mathcal{F}_{\mathrm{HSI}}\left(\mathcal{T}^1_{\mathrm{HSI}}, \mathcal{T}^2_{\mathrm{HSI}}\right) \end{aligned} \tag{8}$$

where $\mathcal{F}_{\mathrm{RGB}}(\cdot)$ and $\mathcal{F}_{\mathrm{HSI}}(\cdot)$ indicate the CD networks in RGB and HSI branch, respectively. $\mathrm{CM}_{\mathrm{RGB}}$ and $\mathrm{CM}_{\mathrm{HSI}}$ are the change maps generated by the networks. It is notable that $\mathcal{F}_{\mathrm{RGB}}(\cdot)$ and $\mathcal{F}_{\mathrm{HSI}}(\cdot)$ only differ in input channel size to suit imagery with different spectral resolution. To use the complementary information from multispectrum data, we propose a triple similarity loss function to synchronously improve the CD performance of different branches, which can be presented as

$$\begin{aligned} \mathcal{L}_{\mathrm{TS}} = &\ \mathcal{L}_{\mathrm{BCE}}(\mathrm{CM}_{\mathrm{RGB}}, \mathrm{CM}_{\mathrm{HSI}}) \\ &+ \mathcal{L}_{\mathrm{BCE}}(\mathrm{CM}_{\mathrm{RGB}}, \mathrm{CM}_{\mathrm{Ref}}) \\ &+ \mathcal{L}_{\mathrm{BCE}}(\mathrm{CM}_{\mathrm{HSI}}, \mathrm{CM}_{\mathrm{Ref}}) \end{aligned} \tag{9}$$

where $\mathcal{L}_{\mathrm{BCE}}(\cdot)$ and $\mathrm{CM}_{\mathrm{Ref}}$ denote the binary cross-entropy loss function and reference change maps, respectively. And $\mathcal{L}_{\mathrm{TS}}$ represents the total loss, triple similarity loss, of the proposed MCCD framework. Notably, the $\mathrm{CM}_{\mathrm{HSI}}$ is the final output change map of MCCD since the HSI branch can detect changes based on more spectral information and outperform $\mathrm{CM}_{\mathrm{RGB}}$ in experiments. Through training in MCCD, the CD performance of the employed network is collaboratively improved. To conclude, the proposed MCCD is expected to fully and stably utilize the real multispectral RS data and enhanced high spectral resolution pseudo-data to improve the CD performance of existing deep learning-based methods.

TABLE II
QUANTITATIVE RESULTS (IN %) OVER COLLABORATIVE LEARNING ON LEVIR DATASET

Methods	Evaluation Metrics			
	F1-Score	**mIoU**	Precision	Recall
FCN	86.96	87.82	88.40	*85.57*
- w/ SKT only	*87.69*(+0.73)	*88.44*(+0.62)	**90.06**	85.45
- w/ MCCD	**87.94**(+0.98)	**88.65**(+0.83)	*89.85*	**86.11**
ResNet	88.45	89.08	90.26	86.70
- w/ SKT only	*89.23*(+0.78)	*89.75*(+0.67)	*90.36*	**88.14**
- w/ MCCD	**89.35**(+0.90)	**89.86**(+0.78)	**91.98**	*86.88*
FC-EF	83.00	84.63	83.36	**82.65**
- w/ SKT only	*83.69*(+0.69)	*85.19*(+0.56)	*86.81*	*80.79*
- w/ MCCD	**83.94**(+0.94)	**85.39**(+0.76)	**87.99**	80.24
FC-Siam-Conc	84.76	86.08	**93.77**	77.32
- w/ SKT only	*87.04*(+2.28)	*87.92*(+1.84)	*92.18*	*82.45*
- w/ MCCD	**87.22**(+2.46)	**88.05**(+1.97)	89.87	**84.73**
FC-Siam-Diff	83.76	85.28	90.75	77.77
- w/ SKT only	*85.06*(+1.30)	*86.31*(+1.03)	*92.74*	*78.54*
- w/ MCCD	**85.42**(+1.66)	**86.60**(+1.32)	**93.20**	**78.83**
SNU-Net	86.43	87.40	*90.38*	82.81
- w/ SKT only	*87.24*(+0.81)	*88.06*(+0.66)	90.11	*84.54*
- w/ MCCD	**88.03**(+1.60)	**88.73**(+1.33)	**90.71**	**85.50**
STA-Net	83.30	84.77	76.69	**91.15**
- w/ SKT only	*84.75*(+1.45)	*86.02*(+1.25)	*86.35*	83.20
- w/ MCCD	**85.07**(+1.77)	**86.28**(+1.51)	**86.81**	*83.39*
BIT	88.48	89.11	**90.35**	86.69
- w/ SKT only	*88.51*(+0.03)	*89.12*(+0.01)	*89.20*	*87.83*
- w/ MCCD	**88.66**(+0.18)	**89.25**(+0.14)	89.06	**88.28**
ChangeFormer	88.54	89.14	87.79	**89.30**
- w/ SKT only	*88.60*(+0.06)	*89.21*(+0.07)	*89.80*	*87.44*
- w/ MCCD	**88.72**(+0.18)	**89.31**(+0.17)	**90.53**	86.97

IV. EXPERIMENTS

In this section, the details of the experimental setup, including the datasets and implementation information, are first introduced. Then, we evaluate the efficiency of the proposed SKT and MCCD with extensive experiments over various state-of-the-art methods and widely used datasets.

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Fig. 7. Visualized results on LEVIR dataset.

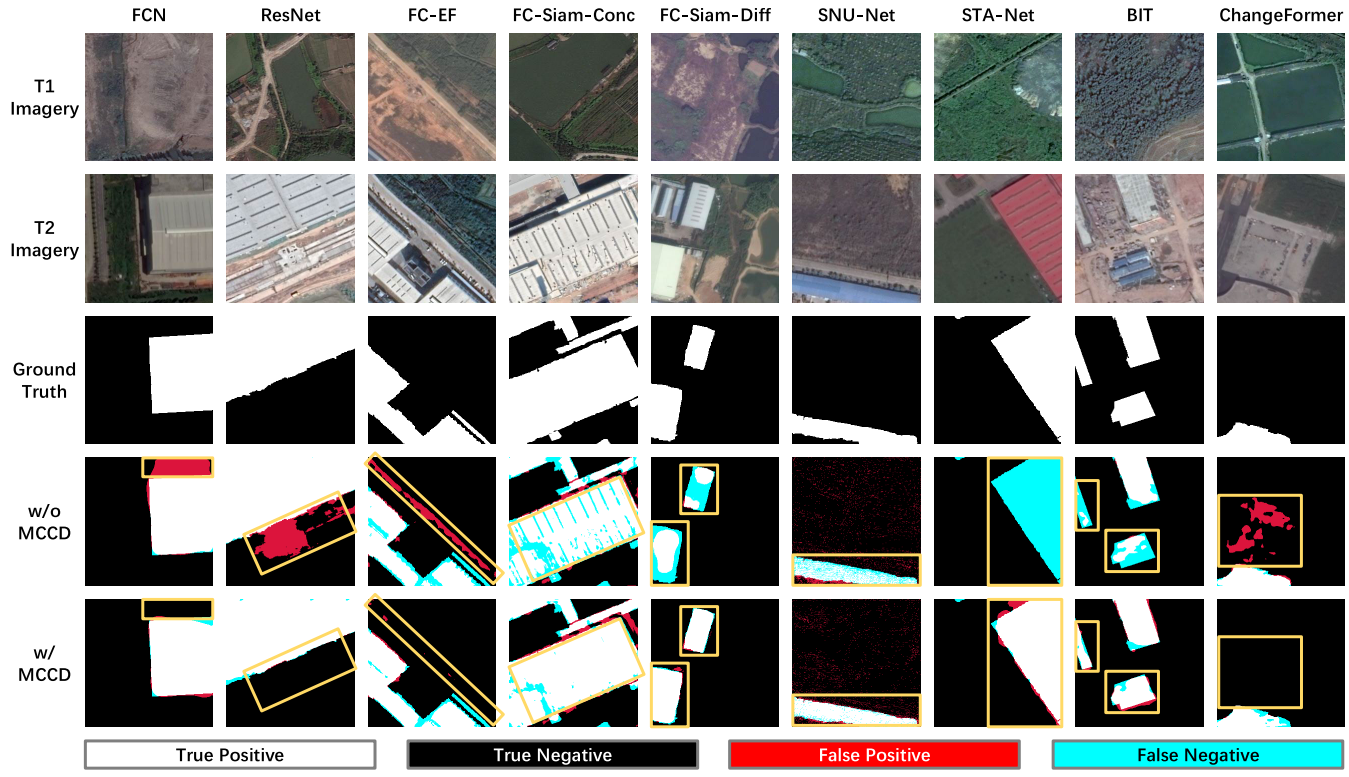


Fig. 8. Visualized results on Google dataset.

A. Experimental Setup

1) Datasets: Three large-scale multispectral CD datasets, i.e., LEarning VIsion RS (LEVIR) [19], Google [67], and WHU [37] datasets, are employed to evaluate our methods. Besides, the CAVE dataset [68] is used to acquire natural spectral knowledge in the proposed SKT. LEVIR dataset has 637 pairs of 1024 × 1024 bi-temporal imagery with a spatial

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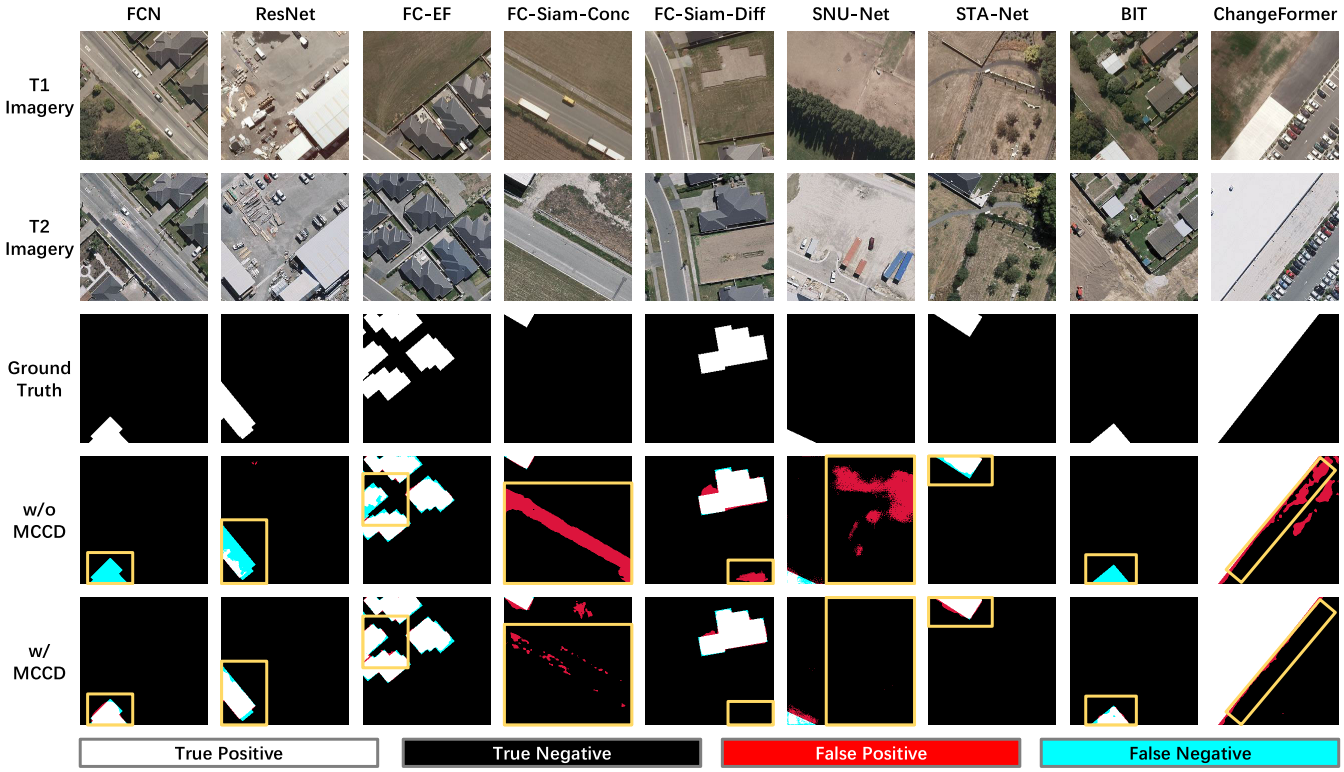


Fig. 9. Visualized results on WHU dataset.

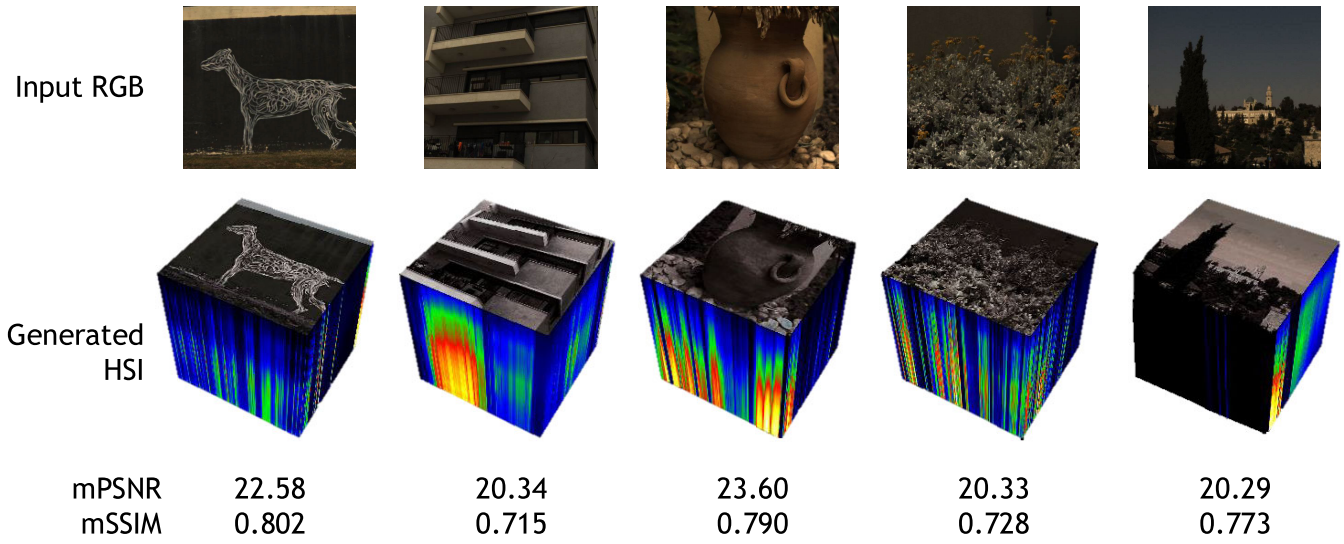


Fig. 10. Visualized results on ARAD dataset. The mPSNR and mSSIM stand for mean peak signal noise ratio and mean structural similarity, respectively.

resolution of 0.5 m, which were obtained by Google Earth API from 2002 to 2018 in Texas, USA. In our experiments, they are resized to 256 × 256 and divided into a training set and validation set, which contain 7120 and 3072 pairs, respectively. Google dataset is composed of 19 season-varying bi-temporal image pairs with a spatial resolution of 0.55 m, covering the suburban areas of Guangzhou city from 2006 to 2019. It is also resized to 256 × 256 image pairs, 2191 pairs for training, and 939 for testing. The WHU dataset was proposed by Wuhan University in 2019, which focuses on the change of building objects around the land with a size of 20.5 km^2 in Christchurch, New Zealand. These images possess a high resolution, with the size of 32 507 × 15 354 pixels and a spatial resolution of 0.075 m. In this work, the WHU dataset is divided into 3480 pairs of image pairs with the size of 256 × 256 pixels; 2396 of them are allocated for training, while the remaining 1084 pairs are used for testing. CAVE dataset is composed of 32 RGB-hyperspectra image pairs which depict different scenes and objects. The natural HSI inside is full spectral resolution reflectance data from 400 to 700 nm at 10 nm steps. Notably, some of the image pairs are not used in the SKT, since there is a slight spatial mismatch over them. Some examples of these datasets are shown in Figs. 5 and 6.

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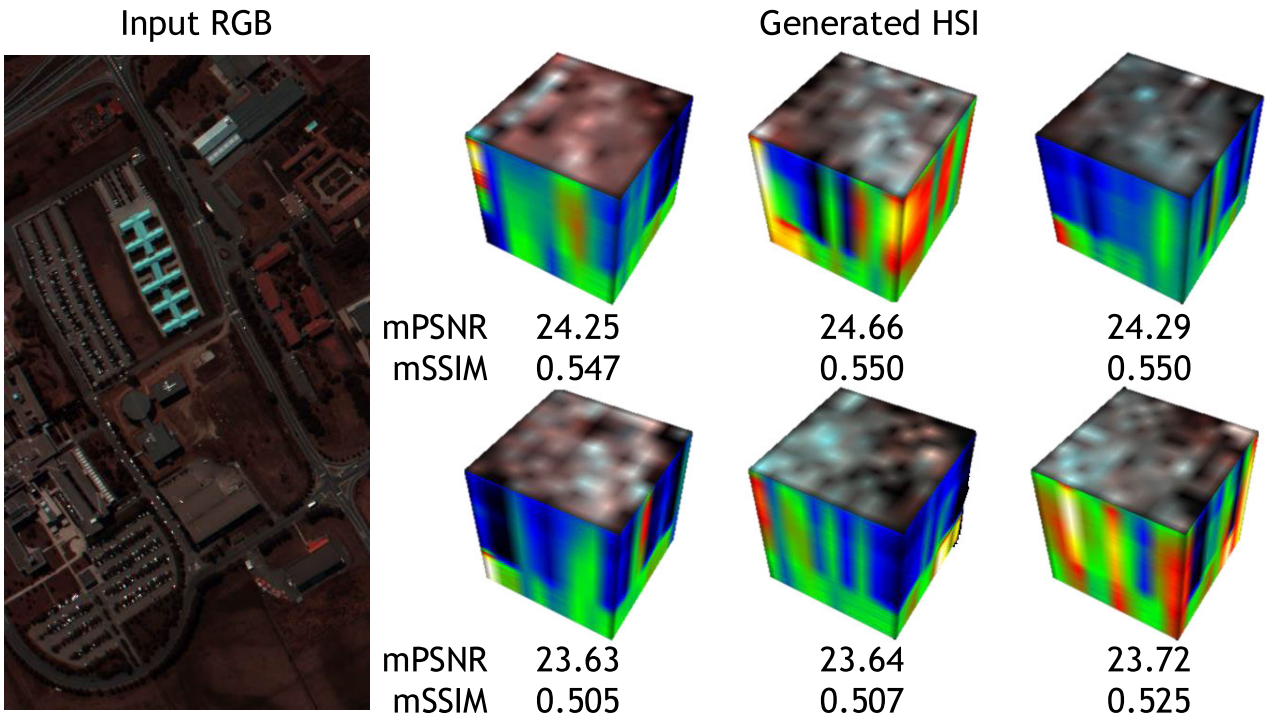


Fig. 11. Visualized results on Pavia University dataset. The mPSNR and mSSIM stand for mean peak signal noise ratio and mean structural similarity, respectively.

TABLE III
QUANTITATIVE RESULTS (IN %) OVER COLLABORATIVE LEARNING ON GOOGLE DATASET

Methods	Evaluation Metrics			
	F1-Score	**mIoU**	Precision	Recall
FCN	78.70	80.09	81.59	**76.00**
- w/ SKT only	*79.54(+0.84)*	*80.79(+0.70)*	*83.71*	75.77
- w/ MCCD	**79.68(+0.98)**	**80.90(+0.81)**	**83.89**	*75.87*
ResNet	*81.21*	*82.01*	80.55	**81.89**
- w/ SKT only	81.05(-0.16)	81.99(-0.67)	*84.79*	77.62
- w/ MCCD	**82.58(+1.37)**	**83.27(+1.26)**	**87.54**	*78.15*
FC-EF	65.96	71.30	77.69	57.31
- w/ SKT only	**68.75(+2.79)**	**73.18(+1.88)**	*82.18*	**59.10**
- w/ MCCD	*68.32(+2.36)*	*72.93(+1.63)*	**83.07**	*58.02*
FC-Siam-Conc	*66.83*	*71.98*	**82.39**	56.21
- w/ SKT only	66.62(-0.21)	71.77(-0.21)	79.62	*57.27*
- w/ MCCD	**67.82(+0.99)**	**72.55(+0.57)**	*80.71*	**58.48**
FC-Siam-Diff	62.33	69.22	*80.29*	*50.94*
- w/ SKT only	**63.82(+1.49)**	**70.30(+1.08)**	**88.65**	49.86
- w/ MCCD	*62.83(+0.50)*	*69.33(+0.11)*	74.78	**54.18**
SNU-Net	49.58	60.84	49.17	**50.01**
- w/ SKT only	*51.21(+1.63)*	*62.77(+1.93)*	**64.76**	42.35
- w/ MCCD	**51.61(+2.03)**	**62.78(+1.94)**	*61.22*	*44.61*
STA-Net	*61.10*	*68.18*	*70.75*	*53.76*
- w/ SKT only	59.00(-2.10)	65.99(-2.19)	55.45	**63.02**
- w/ MCCD	**62.51(+1.41)**	**69.26(+1.08)**	**78.31**	52.01
BIT	78.96	80.46	**88.66**	71.17
- w/ SKT only	*80.55(+1.59)*	*81.63(+1.17)*	86.15	**75.62**
- w/ MCCD	**80.64(+1.68)**	**81.71(+1.25)**	*86.74*	*75.33*
ChangeFormer	78.29	79.89	*85.65*	72.10
- w/ SKT only	*80.32(+2.03)*	*81.49(+1.60)*	**87.85**	*73.97*
- w/ MCCD	**81.49**(+3.20)	**82.35**(+2.46)	85.07	**78.20**

TABLE IV
QUANTITATIVE RESULTS (IN %) OVER SPECTRAL CALIBRATION ON LEVIR DATASET

Methods	Evaluation Metrics			
	F1-Score	**mIoU**	Precision	Recall
FCN	86.96	87.82	88.40	85.57
- w/ MCCD w/o SC	*87.78(+0.82)*	*88.51(+0.69)*	**89.93**	*85.73*
- w/ MCCD	**87.94(+0.98)**	**88.65(+0.83)**	*89.85*	**86.11**
ResNet	88.45	89.08	90.26	86.70
- w/ MCCD w/o SC	*88.98(+0.53)*	*89.53(+0.45)*	*91.34*	*86.73*
- w/ MCCD	**89.35(+0.90)**	**89.86(+0.78)**	**91.98**	**86.88**
FC-EF	*83.00*	*84.63*	83.36	**82.65**
- w/ MCCD w/o SC	81.26(-1.74)	83.36(-1.27)	**88.77**	74.93
- w/ MCCD	**83.94(+0.94)**	**85.39(+0.76)**	*87.99*	*80.24*
FC-Siam-Conc	84.76	86.08	**93.77**	77.32
- w/ MCCD w/o SC	*86.78(+2.02)*	*87.69(+1.61)*	*90.10*	*83.70*
- w/ MCCD	**87.22(+2.46)**	**88.05(+1.97)**	89.87	**84.73**
FC-Siam-Diff	*83.76*	*85.28*	90.75	*77.77*
- w/ MCCD w/o SC	83.42(-0.34)	85.04(-0.24)	**94.00**	74.98
- w/ MCCD	**85.42(+1.66)**	**86.60(+1.32)**	*93.20*	**78.83**
SNU-Net	86.43	87.40	*90.38*	82.81
- w/ MCCD w/o SC	*86.81(+0.38)*	*87.71(+0.31)*	89.29	*84.47*
- w/ MCCD	**88.03(+1.60)**	**88.73(+1.33)**	**90.71**	**85.50**
STA-Net	83.30	84.77	76.69	**91.15**
- w/ MCCD w/o SC	*84.14(+0.84)*	*85.53(+0.76)*	*85.98*	82.37
- w/ MCCD	**85.07(+1.77)**	**86.28(+1.51)**	**86.81**	*83.39*
BIT	*88.48*	*89.11*	**90.35**	*86.69*
- w/ MCCD w/o SC	87.79(-0.69)	88.52(-0.59)	*90.34*	85.37
- w/ MCCD	**88.66(+0.18)**	**89.25(+0.14)**	89.06	**88.28**
ChangeFormer	*88.54*	*89.14*	87.79	**89.30**
- w/ MCCD w/o SC	88.10(-0.44)	88.77(-0.37)	*88.32*	*87.88*
- w/ MCCD	**88.72**(+0.18)	**89.31**(+0.17)	**90.53**	86.97

2) Implementation Details: First, we utilize the proposed encoder–decoder architecture to conduct SKT over a CAVE dataset and a CD dataset. Which CD dataset to use in the spectral calibration depends on which CD dataset to use

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TABLE V
QUANTITATIVE RESULTS (IN %) OVER SPECTRAL CALIBRATION ON GOOGLE DATASET

Methods	Evaluation Metrics			
	F1-Score	**mIoU**	Precision	Recall
FCN	78.70	80.09	81.59	**76.00**
- w/ MCCD w/o SC	*79.09(+0.39)*	*80.44(+0.35)*	*83.62*	75.03
- w/ MCCD	**79.68(+0.98)**	**80.90(+0.81)**	**83.89**	*75.87*
ResNet	81.21	82.01	80.55	**81.89**
- w/ MCCD w/o SC	*81.29(+0.08)*	*82.26(+0.25)*	**88.16**	75.41
- w/ MCCD	**82.58(+1.37)**	**83.27(+1.26)**	*87.54*	*78.15*
FC-EF	65.96	71.30	77.69	57.31
- w/ MCCD w/o SC	*68.18(+2.22)*	*72.82(+1.52)*	*82.28*	**58.21**
- w/ MCCD	**68.32(+2.36)**	**72.93(+1.63)**	**83.07**	*58.02*
FC-Siam-Conc	66.83	71.98	*82.39*	56.21
- w/ MCCD w/o SC	*67.80(+0.97)*	**72.61(+0.63)**	**83.38**	*57.13*
- w/ MCCD	**67.82(+0.99)**	*72.55(+0.57)*	80.71	**58.48**
FC-Siam-Diff	*62.33*	*69.22*	**80.29**	50.94
- w/ MCCD w/o SC	61.58(-0.75)	68.31(-0.91)	67.75	**56.44**
- w/ MCCD	**62.83(+0.50)**	**69.33(+0.11)**	*74.78*	*54.18*
SNU-Net	49.58	60.84	49.17	**50.01**
- w/ MCCD w/o SC	*49.61(+0.03)*	*61.75(+0.91)*	*59.67*	42.46
- w/ MCCD	**51.61(+2.03)**	**62.78(+1.94)**	**61.22**	*44.61*
STA-Net	*61.10*	*68.18*	*70.75*	**53.76**
- w/ MCCD w/o SC	60.45(-0.65)	67.79(-0.39)	70.16	*53.10*
- w/ MCCD	**62.51(+1.41)**	**69.26(+1.08)**	**78.31**	52.01
BIT	78.96	80.46	**88.66**	71.17
- w/ MCCD w/o SC	*80.08(+1.12)*	*81.21(+0.75)*	83.57	**76.88**
- w/ MCCD	**80.64(+1.68)**	**81.71(+1.25)**	*86.74*	*75.33*
ChangeFormer	78.29	79.89	**85.65**	72.10
- w/ MCCD w/o SC	*79.94(+1.65)*	*81.11(+1.22)*	84.00	*76.26*
- w/ MCCD	**81.49**(+3.20)	**82.35**(+2.46)	**85.07**	**78.20**

for CD in the MCCD. In the experiments over MCCD, the hyperparameters of CD networks are set to be consistent with the original works if available. Besides, the hyperspectral branch of MCCD only changes the input channel size of employed CD networks, and the change map generated by the hyperspectral branch is the output of MCCD. All of them are fully trained, and the best models are selected for comparison. The data augmentation strategy is the same for all methods, which is none. All the experiments are implemented over the Pytorch framework on a single NVIDIA 3090 GPU. Four evaluation metrics, i.e., $F1$-Score, mean intersection over union (mIoU), precision rate, and recall rate, are employed for better comparison. With these global evaluation indicators, the performance of these methods can be further evaluated.

B. Evaluation Over State-of-the-Art Baselines

To evaluate the efficiency of proposed MCCD framework with SKT-based pseudo-hyperspectral RS imagery for CD, we conduct the MCCD over several state-of-the-art CD methods and widely used backbone networks, including FCN [14], ResNet [15], FC-EF [16], FC-Siam-Conc [16], FC-Siam-Diff [16], SNU-Net [17], STA-Net [19], BIT [20], and ChangeFormer [69]. Two backbones are introduced to test the versatility and scalability of MCCD since many existing CD methods are built over these two networks. For example, FC-EF, FC-Siam-Conc, and FC-Siam-Diff are based on the architecture of FCN. And STA-Net employs ResNet as its backbone. The corresponding quantitative and visualized results are illustrated in Table I and Figs. 7–9, respectively. In the table, the $w/$ stands for with, the w/o stands for without. The number in parentheses indicates the increase/decrease of evaluation indicators compared to that of the bare networks without any modification. Notably, the baselines with MCCD successfully outperform their origin versions in terms of $F1$-Score and mIoU. It indicates that the proposed MCCD has significant potential to boost the CD performance of existing methods with the help of spectral-enhanced RS data via SKT. For instance, the FC-Siam-Conc with MCCD outperforms its baseline for more than 2% in $F1$-Score. Besides, the visual results demonstrate that MCCD can help the CD methods to gain better recognition of changing land cover, which leads to fewer false alarms and missing alarms. For example, in the first column of Fig. 7, the proposed MCCD can help FCN to detect fewer false positive and false negative pixels at the same time. Moreover, it can be seen from the column of STA-Net in Fig. 8 that the proposed MCCD can significantly lower the false negative pixels, thus improving the performance. On the one hand, richer spectral information may help the methods detect more different information, thus increasing the Recall rate. However, this also increases the potential of false alarms. On the other hand, more spectral information can help the CD methods better identify the required change information, thus increasing the Precision rate. However, this also increases the potential of missing alarming. This may cause the fluctuations of Precision and Recall if using the proposed MCCD. But in the end, the comprehensive CD performance is improved, which can be seen from mIoU and $F1$-Score. It may suggest that higher spectral resolution brought by SKT is able to improve the CD performance for existing methods.

C. Ablation Study

1) Influence of Collaborative Training: In this section, we test the significance of the dual-branch collaborative training process in the proposed MCCD by removing the RGB branch to only use pseudo-HSI in the CD. In principle, learning over the origin multispectral RS data is not that necessary if the pseudo-HSI is well generated without distortion, since an extra branch in the MCCD can be very expensive. To quantify its impact, we only input the pseudo-data generated by SKT into the baselines and train them over LEVIR and Google datasets. The quantitative results are shown in Tables II and III, respectively. From them, it can be concluded that using only pseudo-HSI also brings improvement in CD performance in most cases. However, the performance improvement brought by SKT only is not as great as that of full MCCDfv, some experimental cases in Table III indicate that using the SKT only may cause degradation in the CD performance of the Google dataset. Based on this fact, it indicates that the collaborative training pipeline can leverage multispectrum data to bring more performance improvement and avoid performance degradation. On the other hand, using only SKT to improve CD performance is also acceptable when limited computing resources or time is provided.

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TABLE VI
QUANTITATIVE RESULTS (IN %) OVER MCCD WITH THE SKT TRAINED IN ARAD SPECTRAL SUPER-RESOLUTION DATASET

Methods	LEVIR				Google				WHU			
	F1	**mIOU**	Precision	Recall	**F1**	**mIOU**	Precision	Recall	**F1**	**mIOU**	Precision	Recall
FCN	86.96	87.82	88.40	85.57	78.70	80.09	81.59	76.00	85.18	85.46	88.88	81.78
- w/ MCCD	**88.38**	**89.01**	**89.52**	**87.26**	**81.39**	**82.25**	**84.48**	**78.51**	**89.27**	**89.09**	**90.73**	**87.86**
ResNet	88.45	89.08	**90.26**	86.70	81.21	82.01	80.55	**81.89**	85.66	85.85	87.48	83.93
- w/ MCCD	**89.35**	**89.85**	89.62	**89.08**	**82.15**	**82.82**	**83.01**	81.30	**90.53**	**90.29**	**93.98**	**87.32**

TABLE VII
QUANTITATIVE RESULTS (IN %) OVER MCCD WITH THE SKT TRAINED IN PAVIA UNIVERSITY DATASET

Methods	LEVIR				Google				WHU			
	F1	**mIOU**	Precision	Recall	**F1**	**mIOU**	Precision	Recall	**F1**	**mIOU**	Precision	Recall
FCN	86.96	87.82	88.40	85.57	78.70	80.09	81.59	76.00	85.18	85.46	88.88	81.78
- w/ MCCD	**87.82**	**88.55**	**89.44**	**86.26**	**81.89**	**82.65**	**84.39**	**79.52**	**88.29**	**88.20**	**90.56**	**86.13**
ResNet	88.45	89.08	**90.26**	86.70	81.21	82.01	80.55	**81.89**	85.66	85.85	87.48	83.93
- w/ MCCD	**89.16**	**89.69**	90.20	**88.15**	**81.79**	**82.51**	**82.34**	81.24	**91.43**	**91.13**	**93.16**	**89.76**

2) Influence of Spectral Calibration: In the proposed SKT, we employ spectral calibration to make the learned natural R2H mapping knowledge fit into the manner of RS data, since unhealthy pseudo-HSI may result in the degradation of CD performance. Whether it is necessary is validated by using unregulated pseudo-RS data in the MCCD. The experiments are also carried out over both two CD datasets. The corresponding experimental results are listed in Tables IV and V, respectively, which indicate that the SKT without spectral calibration may degrade CD performance instead of improving it in some cases. From quantitative results, it can be concluded that spectral calibration is necessary for SKT and MCCD frameworks for CD tasks since natural R2H spectral mapping is unsuitable for RS multispectral data which greatly differs from natural optical imagery. To ensure a sound transfer of spectral information and reach higher CD performance, spectral calibration is essential for the proposed SKT framework.

3) Verification of SKT Strategy: In the proposed SKT, we employ a CAVE dataset to build the spectral mapping ability. However, whether this strategy is universally effective for CD when using other spectral super-resolution datasets still waits to be verified. To address this problem, we have added another two datasets, ARAD [70] and Pavia University datasets, which are natural HSI and remotely sensed HSI, respectively, to verify whether this strategy is universally effective for CD. To be specific, we train the SKT over ARAD and Pavia University datasets and utilize it in the proposed MCCD in three large-scale CD datasets. The corresponding experimental results are listed in Tables VI and VII, respectively, which suggest that the proposed MCCD and SKT do not rely on a specific hyperspectral dataset to improve the CD performance. Besides, we conduct extra experiments to evaluate the spectral recovery ability of the proposed SKT over ARAD and Pavia University datasets. The Pavia University dataset is acquired by the ROSIS sensor, with a spatial size of 610 × 340 pixels, spectral bands of 103, and a spatial resolution of 1.3 m. Considering that this dataset is not built for spectral super-resolution, we split it into 2074 10 × 10 nonoverlapping image patches; 70% of these patches are utilized for training and the rest are for validation. According to other works that used this dataset, we select 67-21-11 bands as the R-G-B bands for input. The corresponding results are shown in Figs. 10 and 11. The results suggest that the proposed SKT only has moderately spectral super-resolution ability which may not outperform the state-of-the-art spectral recovery methods. However, the proposed SKT aims to improve the performance of CD methods. Based on this, it is acceptable since the proposed SKT has been proven useful for CD tasks.

V. CONCLUSION

In this work, we invented an SKT strategy, which brings natural hyperspectral information to RS imagery, to increase the spectral resolution of multispectral RS data for better CD performance. It can generate pseudo-HSI from multispectral RS data. Besides, to fully take advantage of bi-temporal pseudo-HSI and real multispectral data, we proposed a new CD framework, MCCD, which collaboratively processes the multispectrum data to further improve the CD performance of existing methods. The proposed SKT and MCCD are flexible enough to be used in a wide range of CD methods. Extensive experimental results over two widely used datasets have demonstrated that they have the potential to significantly improve the CD performance of existing deep learning-based methods. To the best of our knowledge, this is the first work that utilizes natural HSI to construct RS pseudo-HSI and improves CD performance with spectral-enhanced data.

As for future work, more efforts can be made to explore the ability of SKT in other RS downstream tasks, e.g., building detection, and road extraction. The SKT can generate pseudo-HSI for these tasks and potentially improve their performance. Besides, more natural spectral reconstruction datasets can be utilized to facilitate and validate the wide range of applications of pseudo-HSI. Considering that pseudo-HSI

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may have some ineffective spectral bands which may consume additional computing resources, compressive sensing, and similar techniques can be utilized to improve the efficiency.

REFERENCES

[1] Z. Lv, T. Liu, J. A. Benediktsson, and N. Falco, "Land cover change detection techniques: Very-high-resolution optical images: A review," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 1, pp. 44–63, Mar. 2022.

[2] T. Liu et al., "Building change detection for VHR remote sensing images via local–global pyramid network and cross-task transfer learning strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4704817.

[3] T. Liu, M. Gong, F. Jiang, Y. Zhang, and H. Li, "Landslide inventory mapping method based on adaptive histogram-mean distance with bitemporal VHR aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[4] J. Jiang, J. Xiang, E. Yan, Y. Song, and D. Mo, "Forest-CD: Forest change detection network based on VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[5] M. Gong, J. Zhao, J. Liu, Q. Miao, and L. Jiao, "Change detection in synthetic aperture radar images based on deep neural networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 27, no. 1, pp. 125–138, Jan. 2016.

[6] F. Liu, L. Jiao, X. Tang, S. Yang, W. Ma, and B. Hou, "Local restricted convolutional neural network for change detection in polarimetric SAR images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 3, pp. 818–833, Mar. 2019.

[7] H. Zhang, M. Lin, G. Yang, and L. Zhang, "ESCNet: An end-to-end superpixel-enhanced change detection network for very-high-resolution remote sensing images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 1, pp. 28–42, Jan. 2023.

[8] Y. Wu, J. Li, Y. Yuan, A. K. Qin, Q.-G. Miao, and M.-G. Gong, "Commonality autoencoder: Learning common features for change detection from heterogeneous images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 9, pp. 4257–4270, Sep. 2022.

[9] J. Liu, M. Gong, K. Qin, and P. Zhang, "A deep convolutional coupling network for change detection based on heterogeneous optical and radar images," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 3, pp. 545–559, Mar. 2018.

[10] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[11] Q. Zhu, X. Sun, Y. Zhong, and L. Zhang, "High-resolution remote sensing image scene understanding: A review," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 3061–3064.

[12] L. Khelifi and M. Mignotte, "Deep learning for change detection in remote sensing images: Comprehensive review and meta-analysis," *IEEE Access*, vol. 8, pp. 126385–126400, 2020.

[13] F. Luo, T. Zhou, J. Liu, T. Guo, X. Gong, and J. Ren, "Multiscale diff-changed feature fusion network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5502713.

[14] J. Long, E. Shelhamer, and T. Darrell, "Fully convolutional networks for semantic segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3431–3440.

[15] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[16] R. Caye Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[17] S. Fang, K. Li, J. Shao, and Z. Li, "SNUNet-CD: A densely connected Siamese network for change detection of VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[18] K. Han et al., "A survey on vision transformer," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 1, pp. 87–110, Jan. 2023.

[19] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[20] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5607514.

[21] X. Sun, L. Zhang, H. Yang, T. Wu, Y. Cen, and Y. Guo, "Enhancement of spectral resolution for remotely sensed multispectral image," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 5, pp. 2198–2211, May 2015.

[22] L. Liu, D. Hong, L. Ni, and L. Gao, "Multilayer cascade screening strategy for semi-supervised change detection in hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1926–1940, 2022.

[23] H. Jiang et al., "A survey on deep learning-based change detection from high-resolution remote sensing images," *Remote Sens.*, vol. 14, no. 7, p. 1552, Mar. 2022.

[24] Z. Zheng, A. Ma, L. Zhang, and Y. Zhong, "Change is everywhere: Single-temporal supervised object change detection in remote sensing imagery," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 15173–15182.

[25] M. Seo, H. Lee, Y. Jeon, and J. Seo, "Self-pair: Synthesizing changes from single source for object change detection in remote sensing imagery," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 6363–6372.

[26] S. Mishra, P. Shrivastava, and P. Dhurvey, "Change detection techniques in remote sensing: A review," *Int. J. Wireless Mobile Commun. Ind. Syst.*, vol. 4, no. 1, pp. 1–8, 2017.

[27] G. Liu, L. Jiao, F. Liu, H. Zhong, and S. Wang, "A new patch based change detector for polarimetric SAR data," *Pattern Recognit.*, vol. 48, no. 3, pp. 685–695, Mar. 2015.

[28] Y. Sun, L. Lei, X. Li, H. Sun, and G. Kuang, "Nonlocal patch similarity based heterogeneous remote sensing change detection," *Pattern Recognit.*, vol. 109, Jan. 2021, Art. no. 107598.

[29] Y. Shu, W. Li, M. Yang, P. Cheng, and S. Han, "Patch-based change detection method for SAR images with label updating strategy," *Remote Sens.*, vol. 13, no. 7, p. 1236, Mar. 2021.

[30] Y. Lei, X. Liu, J. Shi, C. Lei, and J. Wang, "Multiscale superpixel segmentation with deep features for change detection," *IEEE Access*, vol. 7, pp. 36600–36616, 2019.

[31] X. Zhang et al., "Two-phase object-based deep learning for multitemporal SAR image change detection," *Remote Sens.*, vol. 12, no. 3, p. 548, Feb. 2020.

[32] X. Zheng, X. Chen, X. Lu, and B. Sun, "Unsupervised change detection by cross-resolution difference learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5606616.

[33] N. Lv, C. Chen, T. Qiu, and A. K. Sangaiah, "Deep learning and superpixel feature extraction based on contractive autoencoder for change detection in SAR images," *IEEE Trans. Ind. Informat.*, vol. 14, no. 12, pp. 5530–5538, Dec. 2018.

[34] M. Gong, T. Zhan, P. Zhang, and Q. Miao, "Superpixel-based difference representation learning for change detection in multispectral remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 5, pp. 2658–2673, May 2017.

[35] P. Zhang, M. Gong, L. Su, J. Liu, and Z. Li, "Change detection based on deep feature representation and mapping transformation for multi-spatial-resolution remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 116, pp. 24–41, Jun. 2016.

[36] Y. Wang et al., "Mask DeepLab: End-to-end image segmentation for change detection in high-resolution remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 104, Dec. 2021, Art. no. 102582.

[37] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[38] Q. Shi, M. Liu, S. Li, X. Liu, F. Wang, and L. Zhang, "A deeply supervised attention metric-based network and an open aerial image dataset for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5604816.

[39] M. A. Lebedev, Y. V. Vizilter, O. V. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vol. 2, pp. 565–571, May 2018.

[40] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, no. 7553, pp. 436–444, 2015.

[41] Y. Guo, Y. Liu, A. Oerlemans, S. Lao, S. Wu, and M. S. Lew, "Deep learning for visual understanding: A review," *Neurocomputing*, vol. 187, pp. 27–48, Apr. 2016.

[42] X. X. Zhu et al., "Deep learning in remote sensing: A comprehensive review and list of resources," *IEEE Geosci. Remote Sens. Mag.*, vol. 5, no. 4, pp. 8–36, Dec. 2017.

[43] G. Cheng, X. Xie, J. Han, L. Guo, and G.-S. Xia, "Remote sensing image scene classification meets deep learning: Challenges, methods, benchmarks, and opportunities," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3735–3756, 2020.

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[44] X. Yuan, J. Shi, and L. Gu, "A review of deep learning methods for semantic segmentation of remote sensing imagery," *Expert Syst. Appl.*, vol. 169, May 2021, Art. no. 114417.

[45] A. Shafique, G. Cao, Z. Khan, M. Asad, and M. Aslam, "Deep learning-based change detection in remote sensing images: A review," *Remote Sens.*, vol. 14, no. 4, p. 871, Feb. 2022.

[46] T. Chen, Z. Lu, Y. Yang, Y. Zhang, B. Du, and A. Plaza, "A Siamese network based U-Net for change detection in high resolution remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 2357–2369, 2022.

[47] F. Song, R. Ma, T. Lei, and Z. Peng, "RAIH-Det: An end-to-end rotated aircraft and aircraft head detector based on ConvNeXt and cyclical focal loss in optical remote sensing images," *Remote Sens.*, vol. 15, no. 9, p. 2364, Apr. 2023.

[48] H. Zheng et al., "HFA-net: High frequency attention Siamese network for building change detection in VHR remote sensing images," *Pattern Recognit.*, vol. 129, Sep. 2022, Art. no. 108717.

[49] T. Lei et al., "Difference enhancement and spatial–spectral nonlocal network for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4507013.

[50] N. Wang, W. Li, R. Tao, and Q. Du, "Graph-based block-level urban change detection using Sentinel-2 time series," *Remote Sens. Environ.*, vol. 274, Jun. 2022, Art. no. 112993.

[51] Q. Zhu et al., "Land-Use/Land-Cover change detection based on a Siamese global learning framework for high spatial resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 63–78, Feb. 2022.

[52] M.-H. Guo et al., "Attention mechanisms in computer vision: A survey," *Comput. Vis. Media*, vol. 8, pp. 331–368, Mar. 2022.

[53] Z. Niu, G. Zhong, and H. Yu, "A review on the attention mechanism of deep learning," *Neurocomputing*, vol. 452, pp. 48–62, Sep. 2021.

[54] F. Wang et al., "Residual attention network for image classification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6450–6458.

[55] M. Zhang, H. Zheng, M. Gong, Y. Wu, H. Li, and X. Jiang, "Self-structured pyramid network with parallel spatial-channel attention for change detection in VHR remote sensed imagery," *Pattern Recognit.*, vol. 138, Jun. 2023, Art. no. 109354.

[56] Y. Zhou, Z. Chen, P. Li, H. Song, C. L. P. Chen, and B. Sheng, "FSAD-Net: Feedback spatial attention dehazing network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 10, pp. 7719–7733, Oct. 2022.

[57] H.-C. Li, W.-S. Hu, W. Li, J. Li, Q. Du, and A. Plaza, "A3 CLNN: Spatial, spectral and multiscale attention ConvLSTM neural network for multisource remote sensing data classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 2, pp. 747–761, Feb. 2022.

[58] H. Tan, X. Liu, B. Yin, and X. Li, "MHSA-Net: Multihead self-attention network for occluded person re-identification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 11, pp. 8210–8224, Nov. 2022.

[59] L. Mou and X. X. Zhu, "Learning to pay attention on spectral domain: A spectral attention module-based convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 110–122, Jan. 2020.

[60] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.

[61] L. Wang, L. Wang, Q. Wang, and P. M. Atkinson, "SSA-SiamNet: Spectral–spatial-Wise attention-based Siamese network for hyperspectral image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5510018.

[62] P. V. Arun, K. M. Buddhiraju, A. Porwal, and J. Chanussot, "CNN based spectral super-resolution of remote sensing images," *Signal Process.*, vol. 169, Apr. 2020, Art. no. 107394.

[63] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5509417.

[64] L. Gao, D. Hong, J. Yao, B. Zhang, P. Gamba, and J. Chanussot, "Spectral superresolution of multispectral imagery with joint sparse and low-rank learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2269–2280, Mar. 2021.

[65] J. He, J. Li, Q. Yuan, H. Shen, and L. Zhang, "Spectral response function-guided deep optimization-driven network for spectral super-resolution," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 9, pp. 4213–4227, Sep. 2022.

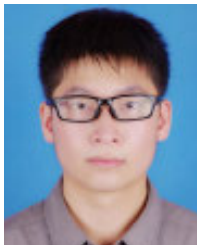
[66] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2242–2251.

[67] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, H. Ding, and X. Huang, "SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5891–5906, Jul. 2021.

[68] F. Yasuma, T. Mitsunaga, D. Iso, and S. K. Nayar, "Generalized assorted pixel camera: Postcapture control of resolution, dynamic range, and spectrum," *IEEE Trans. Image Process.*, vol. 19, no. 9, pp. 2241–2253, Sep. 2010.

[69] W. G. C. Bandara and V. M. Patel, "A transformer-based Siamese network for change detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 207–210.

[70] B. Arad et al., "NTIRE 2022 spectral recovery challenge and data set," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2022, pp. 862–880.



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